

Biology, 2016 EC, model exam, Arada

Grade 12 · Biology · 2016 E.C.

Time allowed: 120 minutes

Questions: 95

Total marks: 95

Name _____ Grade _____ ID No. _____ Date _____

1. Which one of the following term and its definition is not correct? (1)
 - A. Hypothesis→ statement that explain observed facts and predicts new outcomes
 - B. Journal → special magazine were scientists research published
 - C. Peer review → the use of scientific research to meet human desires
 - D. Scientific method →the process by which scientists approach their work
2. Which one of the following disease is not correctly matched with its causative agent? (1)
 - A. Tuberculosis→Mycobacterium tuberculosis
 - B. Gastroenteritis →Rota viruses
 - C. Cholera → Vibrio cholera
 - D. Chancroid→ Trepomema pallidum
3. What is the field of Ethiopian biologist Tewolde Berhan Gebre Egzabher? (1)
 - A. Marine biology
 - B. Conservation biology
 - C. Genetics
 - D. Microbiology
4. What is the purpose of using antiseptics in controlling microorganisms? (1)
 - A. To inhibit microbial growth in food products
 - B. To control microorganisms on inanimate surfaces
 - C. To sterilize equipments and media
 - D. To control microorganisms on living tissue
5. What is the foremost goal of gene therapy in modern biotechnology? (1)
 - A. To cure genetic diseases by replacing defective gene
 - B. To produce vaccine against infectious disease
 - C. To create genetically modified organism
 - D. To produce cloned organism
6. What is the role of enzymes in traditional biotechnology? (1)
 - A. To encode specific trait in an organism
 - B. To catalyze biochemical reaction of in fermentation
 - C. To modify gene of an organism
 - D. To cut DNA into specific gene
7. If pea plant with the genotype Rr is crossed with rr genotype, what is the phenotypic ratio of their offspring? (1)
 - A. 1:1
 - B. 2:1
 - C. 3:1
 - D. 1:2:1

-
- 8.** Mendel's law of segregation states that:- (1)
- A. Alleles of different genes assort independently during meiosis
 - B. Alleles of the same gene separated into different during meiosis
 - C. The expression of one gene influenced by expression of other gene
 - D. Genes located on the same chromosome are inherited together
-
- 9.** Which of the following nervous system is responsible for controlling involuntary action such as heart rate? (1)
- A. Central nervous system
 - B. Peripheral nervous system
 - C. Autonomous nervous system
 - D. Somatic nervous system
-
- 10.** Which part of the human eye control the amount of light enters into the pupil? (1)
- A. Retina
 - B. Lens
 - C. Iris
 - D. Cornea
-
- 11.** Which part of human ear is responsible for maintaining balance and equilibrium? (1)
- A. Cochlea
 - B. Semicircular canals
 - C. Eardrum
 - D. Auditory nerve
-
- 12.** Which gland produces insulin and glucagon responsible for controlling blood sugar level? (1)
- A. Thyroid gland
 - B. Adrenal gland
 - C. Pituitary gland
 - D. Pancreas gland
-
- 13.** Which of the following statement correctly explain homeostasis? (1)
- A. Dilation of blood vessels in cold temperature
 - B. Decrease heart rate during physical exercise
 - C. Constrict blood vessels in hot temperature
 - D. Decrease breathing rate during sleep
-
- 14.** What is the purpose of conducting experiments in the scientific method? (1)
- A. To gather data and evidence
 - B. To confirm personal trust
 - C. To prove a hypothesis
 - D. To support predetermined idea
-
- 15.** What is a key threat of biodiversity in Ethiopia? (1)
- A. Strict conservation policies
 - B. Abundant protected areas
 - C. Large scale reforestation attempt
 - D. Climate change and desertification
-
- 16.** How do organisms acquire adaptation over time? (1)
- A. Through direct manipulation of their environment
 - B. Through artificial manipulation of their genes
 - C. Through random mutation of their genes
 - D. Through conscious decision making
-
- 17.** Which of the following describes a structural adaptation in plants? (1)
- A. Hibernate to survive cold season
 - B. Shedding leaves to conserve water during drought
 - C. Camouflage to avoid predator
 - D. Produce energy by photosynthesis

18. All of the following explain the contribution of reforestation to conserving natural resource in Ethiopia except? (1)

- A. Decrease carbon emission
B. Decrease wildlife
C. Decrease desertification
D. Decrease soil erosion

19. What is the term that describes plant response towards touch or physical contact? (1)

- A.** Thigmotropism **B.** Hydrotropism
C. Phototropism **D.** Chemotropism

20. Which of the following plant hormone is responsible inducing seed dormancy? (1)

- A.** Gibberellins **B.** Auxin
C. Absciscic acid **D.** Cytokinin

21. Which of the following alternative contains correct list of cell membrane proteins? (1)

[Figure: The diagram is preceded by the title 'Question n\underline{o} 21 and 22are based on the following figure'. It illustrates the fluid mosaic model of a cell membrane with a phospholipid bilayer. Four components are indicated by labeled boxes: - A points to a large transmembrane protein (integral protein) spanning the bilayer. - B points to a carbohydrate chain attached directly to a phospholipid on the external surface (glycolipid). - C points to a carbohydrate chain attached to a membrane protein on the external surface (glycoprotein). - D points to a protein situated entirely on the intracellular surface of the bilayer (peripheral protein).]

- A. A: glycoprotein and B: peripheral protein, C: integral protein and D: glycolipid,
B. A: integral protein, B: glycolipid, C: peripheral protein and D: glycotien
C. A: integral protein, B: glycolipid, C: glycoprotein and D: peripheral protein
D. A: glycolipid, B: glycoprotein, C: integral protein and D: integral protein

22. Type of membrane protein on the cell surface which plays a key role in the body's immune response is represented by what letter? (1)

[Figure: The diagram is preceded by the title 'Question n\nd 21 and 22 are based on the following figure'. It illustrates the fluid mosaic model of a cell membrane with a phospholipid bilayer. Four components are indicated by labeled boxes: - A points to a large transmembrane protein (integral protein) spanning the bilayer. - B points to a carbohydrate chain attached directly to a phospholipid on the external surface (glycolipid). - C points to a carbohydrate chain attached to a membrane protein on the external surface (glycoprotein). - D points to a protein situated entirely on the intracellular surface of the bilayer (peripheral protein).]

- A. D** **B. A**
C. B **D. C**

23. What is the role of ethylene hormone in plant growth and development? (1)

- A.** Initiate seed germination **B.** Ripening fruit
C. Induce seed dormancy **D.** Elongate cells

24. Which of the following does not describe acquired (adaptive) immune system? (1)

- A. Relies on mechanisms that exist before infection
- B. Relies on mechanisms that adapt after infection
- C. Second line of response
- D. Handled by T- and B- lymphocytes

-
- 25.** Which of the following is not correct about anaerobic respiration /fermentation/ in animals? (1)
- A. Cramping appear due to lack of the ATP to pump Ca^{++} ions back that enable muscle relax
 - B. The accumulation of lactate in a muscle fiber makes the cytoplasm more acidic
 - C. During fermentation, glucose is broken down to lactate
 - D. People who train accumulate more lactate than those who do not train
-
- 26.** In all the following condition diffusion is faster except; (1)
- A. With higher concentration difference
 - B. For smaller molecules
 - C. At high temperature
 - D. Over long distance
-
- 27.** Which one of the following animal possess pouch for completing the development of the embryo? (1)
- A. Placental
 - B. Marsupial
 - C. Egg laying
 - D. Non placental
-
- 28.** Which of the following factor that affect photosynthesis is not correct? (1)
- A. Light intensity limit light dependent reaction by reducing e^- in chlorophyll
 - B. Temperature limit enzyme action in photosynthesis
 - C. Oxygen limit the amount of carbohydrate produced in photosynthesis
 - D. Carbon dioxide limit the initial reaction of light independent
-
- 29.** Question n° 31 is based on the given table below (1)
- Table 1 Tonicity of solution
- Based on the above table, which of the following is correct?
- [Figure: Table 1 Tonicity of solution, consisting of two columns: 'Solution' and 'Cell behavior when placed in solution'. Row 1: Solution 'A', Cell behavior 'Cell swells'. Row 2: Solution 'B', Cell behavior 'Cell size doesn't change'. Row 3: Solution 'C', Cell behavior 'Cell shrinks'.]*
- A. Solution A is hypotonic, solution B is hypertonic and solution C is isotonic
 - B. Solution A is isotonic, solution B is hypotonic and hypertonic and solution
 - C. Solution A is hypotonic, solution B is isotonic and solution C is hypertonic
 - D. Solution A is hypertonic, solution B is hypotonic and solution C is isotonic
-
- 30.** From the following which one is a biotic component of the ecosystem? (1)
- A. Parasite
 - B. Competition
 - C. Disease
 - D. Oxygen
-
- 31.** Which of the following properties of enzymes makes it possible that a single enzyme molecule can act on many substrate molecules? (1)
- A. Enzymes lower the energy of activation
 - B. Enzymes are reused over and over again
 - C. Enzymes are proteins
 - D. Enzymes are substrate specific
-
- 32.** In which type of solution is the water potential more negative than in the cells? (1)
- A. Hypotonic
 - B. Hypertonic
 - C. Isotonic
 - D. Equivalent solution
-
- 33.** Which of the following blood vessels contain valves to prevent backflow of blood? (1)
- A. Capillaries
 - B. Arteries
 - C. Veins
 - D. Venules

-
- 34.** All of the following produce by yeast during fermentation except; (1)
- | | |
|------------|-------------------|
| A. Lactate | B. Carbon dioxide |
| C. ATP | D. Alcohol |
-
- 35.** The scientific name of a house fly is: (1)
- | | |
|---------------------------|----------------------------|
| A. <i>Musca domestica</i> | B. <i>Canis familiaris</i> |
| C. <i>Zea mays</i> | D. <i>Felis domesticus</i> |
-
- 36.** What is the function of mucus in the respiratory system? (1)
- | | |
|------------------------|------------------|
| A. Transporting oxygen | B. Produce sound |
| C. Remove debris | D. Exchange gas |
-
- 37.** Which of the following is not correct in the mechanism of normal breathing in or inhalation? (1)
- A. The intercostal muscle between ribs relax
- B. The volume of thorax increases
- C. Ribs move up and out
- D. Diaphragm muscle contract
-
- 38.** If a sample fruit juice and some drops of Benedict's solution obtain a brick – red precipitate up on warming the mixture, what does the juice contain? (1)
- | | |
|------------|-------------------|
| A. Starch | B. Protein |
| C. Sucrose | D. Reducing sugar |
-
- 39.** Assume the amino acid coding region in mRNA is 1800 nucleotides long, how long is the protein in terms of amino acid number? (1)
- | | |
|---------------------|--------------------|
| A. 1200 amino acids | B. 400 amino acids |
| C. 600 amino acids | D. 300 amino acids |
-
- 40.** At which level of structural organization do proteins have the alpha-helix shape? (1)
- | | |
|-----------------------|-------------------------|
| A. Primary structure | B. Secondary structure |
| C. Tertiary structure | D. Quaternary structure |
-
- 41.** When an enzyme is denatured by heat or extreme pH, which one of the following does it lose? (1)
- | | |
|-------------------------|-----------------------|
| A. Quaternary structure | B. Primary structure |
| C. Secondary structure | D. Tertiary structure |
-
- 42.** Which of the following taxonomic rank is broader than order but narrower than phylum? (1)
- | | |
|------------|-----------|
| A. Kingdom | B. Class |
| C. Phylum | D. Family |
-
- 43.** Who is considered as a father of modern taxonomy? (1)
- | | |
|---------------------|------------------|
| A. Charles Darwin | B. Gregor Mendel |
| C. Carolus Linnaeus | D. Louis Pasteur |

-
- 44.** Which of the following taxonomic rank is correct from largest taxa to smallest? (1)
- A. Kingdom → phylum → class → family → order → genus → species
 - B. Kingdom → phylum → class → order → family → genus → species
 - C. Kingdom → phylum → order → class → family → genus → species
 - D. Kingdom → order → phylum → class → family → genus → species
-
- 45.** Which one of the following enzyme and its function is not correct? (1)
- A. Ligase → joining of nucleotide to form DNA
 - B. Catalase → breaking down peroxide
 - C. Pepsin → digesting proteins
 - D. Lipase → digesting lipids
-
- 46.** What are the environmental advantages of using enzymes in industry? (1)
- A. It helps high production with supply of more heat and emission of more CO₂
 - B. It makes high production possible with less input of heat
 - C. It makes high production possible with high input of heat
 - D. It makes high production possible with emission of more CO₂
-
- 47.** Biological tool that is used to separate components of different molecular weight is known as; (1)
- A. Transect
 - B. GPS
 - C. Quadra
 - D. Centrifuge
-
- 48.** Identify the illegal drug that is known for its strong hallucinogen feature (1)
- A. Cannabis
 - B. Heroin
 - C. Lysergic acid diethyl amide
 - D. Cocaine
-
- 49.** Choose the correct letter that contains the correct order of ossicles found in middle ear from the biggest to the smallest. (1)
- A. Hammer, incus and stapes
 - B. Stapes, hammer and incus
 - C. Incus, hammer and stapes
 - D. Hammer, stapes and incus
-
- 50.** In which of the following mitosis cell division occurs slowly? (1)
- A. RBC
 - B. Nerve cells
 - C. Skin cells
 - D. WBC
-
- 51.** What is the role of hypodermis skin layer? (1)
- A. Contain sensory receptors and help in thermoregulation
 - B. Act as waterproof and prevent entry of pathogens
 - C. Act as energy store and insulation layer
 - D. Act as energy store and prevent entry of pathogens
-
- 52.** If FSH produced by pituitary gland, what will happen to female gonad? (1)
- A. The ova become developed and oestrogen produced
 - B. The ova become developed and progesterone produced
 - C. The ova released from ovary and oestrogen produced
 - D. The ova released from ovary and progesterone produced

53. Which of the following is not physiological method of controlling temperature when temperature goes up in homoeothermis? (1)
- A. Vasodilation
B. Piloerection
C. Painting
D. Licking
54. As compared to saturated fat, unsaturated fat contains less? (1)
- A. Fatty acid
B. Glycerol
C. Hydrogen
D. Oxygen
55. Which of the following is not describes why biodiversity loss is concerned at the moment? (1)
- A. Food insecurity
B. Worsening social interaction
C. Health problem
D. Food security
56. Which of the following is not considered as characteristics of inbreeding? (1)
- A. Increase homozygosity
B. Reduce disease resistance
C. Reduce fertility
D. Increase disease resistance
57. 61. In the Miller and Urey simulation experiment to study organic molecules formation in the primeval atmosphere of the earth all of the following gases were used except : (1)
- A. ammonia
B. oxygen
C. methane
D. nitrogen
58. 62. Rabbits ear can be either short or floppy where short ears are dominant over floppy ears, in the populations of rabbits there are 653 individuals ,104 rabbits have floppy ears and 549 have short ears.find the frequency of dominant allele(P) and recessive allele (Q) in the population respectively? (1)
- A. $P=0.36$ and $Q=0.16$
B. $P=0.6$ and $Q=0.4$
C. $P=0.16$ and $Q=0.6$
D. $P=.36$ and $Q=0.4$
59. 63. Which of the following combination is known as Neo Darwinism? (1)
- A. natural selection and Mendel law of inheritance
B. Natural selection and acquired characters
C. Acquired characters and Mendel law of inheritance
D. Natural selection and Lamarck theory of evolution
60. 64.Which of the following term is used for fungal infection of the nails? (1)
- A. Tineabarbae
B. Tineaungum
C. Tineapedis
D. Tineacorporis
61. 65. Which of the following bacteria produces acetic acid and does not cause food poisoning? (1)
- A. Staphylococcus aureus
B. Acetobacteraceti
C. Bacillus cereus
D. Campylobacter jeluni
62. 66.Which of the following consists of only spirit products? (1)
- A. Whisky, gin ,brandy and vodka
B. Gin ,brandy, beer and whisky
C. Wine , brandy ,gin and vodka
D. Wine ,beer, gin and vodka

63. 67. Yoghurt is made by adding to milk microbes known as (1)
- | | |
|------------------------------------|--------------------------------------|
| A. <i>Fusarium venenatum</i> | B. <i>Streptococcus thermophilus</i> |
| C. <i>Lactobacillus bulgaricus</i> | D. B and C |
-
64. 68. Which of the following is not external part of the leaf? (1)
- | | |
|-----------|---------------------|
| A. Midrib | B. Petiole |
| C. Lamina | D. Mesophyll tissue |
-
65. 69. Which of the following absorb different wavelength of light? (1)
- | | |
|--------------|-------------|
| A. Pigments | B. Enzymes |
| C. Nutrients | D. Minerals |
-
66. 70. Which part of plant leaf contain large number of chloroplasts that carry out most of the photosynthetic activity? (1)
- | | |
|-----------------------|---------------------|
| A. Palisade mesophyll | B. Spongy mesophyll |
| C. Guard cells | D. Epidermal cells |
-
67. 71. Which of the following is not a photoautotroph? (1)
- | | |
|-----------------------------|------------------------|
| A. Purple sulphur bacteria | B. Cyanobacteria |
| C. Nitrogen fixing bacteria | D. Multicellular algae |
-
68. 72. Which part of white light is very important to drive photosynthesis mostly? (1)
- | | |
|----------|-----------|
| A. Green | B. Yellow |
| C. Blue | D. Orange |
-
69. 73. Which of the following is produced during light dependent reaction of photosynthesis? (1)
- | | |
|----------------------------|------------------------------|
| A. glucose, NADPH and ATP | B. oxygen, NADPH and ATP |
| C. ATP, glucose and oxygen | D. NADPH, glucose and oxygen |
-
70. 74. In the calvin cycle which of the following compound is produced in C_3 plants first? (1)
- | | |
|-------------------------------|-------------------------------|
| A. Glyceraldehyde-3-phosphate | B. Glucose -6-phosphate |
| C. Fructose 6 phosphate | D. Glucose, 1, 6 bisphosphate |
-
71. 75. Which of the following comparison of anabolic & catabolic path way is not correct? (1)
- [Figure: A table with two columns titled 'Anabolism' and 'Catabolism', containing four rows labeled A, B, C, and D:*
- Row A: under Anabolism, 'A. Simple substance joined molecules to form simple substances'; under Catabolism, 'A. Complex macromolecules broken down Together to form macro'
 - Row B: under Anabolism, 'B. The process release energy'; under Catabolism, 'B. The process absorb energy'
 - Row C: under Anabolism, 'C. occur in photosynthesis'; under Catabolism, 'C. Occur in respiration'
 - Row D: under Anabolism, 'D. Construction process of metabolism'; under Catabolism, 'D. destruction phase of metabolism'
- | |
|---|
| A. Anabolism: Simple substance joined molecules to form simple substances Catabolism: - Complex macromolecules broken down Together to form macro |
| B. Anabolism: The process release energy Catabolism: - The process absorb energy |
| C. Anabolism: occur in photosynthesis Catabolism: - Occur in respiration |
| D. Anabolism: Construction process of metabolism Catabolism: - destruction phase of metabolism |

-
- 72.** 76. From which of the following gene mutations does sickle cell anemia result? (1)
- | | |
|-----------------|--------------|
| A. Substitution | B. Addition |
| C. Deletion | D. Inversion |
-
- 73.** 77. Which of the following organism add flavor by fermenting meats such a salami, summer sausage & Lebanon bologna? (1)
- | | |
|------------------------------------|--------------------------------------|
| A. <i>Saccharomyces cerevisiae</i> | B. <i>Streptococcus cremoris</i> |
| C. <i>Lactobacillus plantarium</i> | D. <i>Methylophilusmethylophilus</i> |
-
- 74.** 78. Which of the following involve extracting and obtaining energy from their environment by enzyme mediated chemical reaction? (1)
- | | |
|-------------|------------------|
| A. Sunlight | B. Soil |
| C. Water | D. living things |
-
- 75.** Which one of the following does not involve the practical applications of transgenic technology in livestock? (1)
- A. increasing feed utilization & growth rate
 - B. Improving milk production & composition
 - C. Increasing environmental degradation
 - D. Increasing disease resistance in animals
-
- 76.** Which of the following stages of aerobic respiration does not produce ATP? (1)
- | | |
|---------------------------------|----------------------|
| A. The electron transport chain | B. Glycolysis |
| C. The kreb cycle | D. The link reaction |
-
- 77.** Identify the alternatives that contain a pair of energy storage compounds of carbohydrate: (1)
- | | |
|-------------------------|-----------------------|
| A. Cellulose & Starch | B. Starch & Glycogen |
| C. Cellulose & Glycogen | D. Chitin & Cellulose |
-
- 78.** Which of the following process does not require energy from ATP to occur? (1)
- | | |
|----------------------------------|--------------------------------------|
| A. Transmission of nerve impulse | B. Initial reaction of respiration |
| C. synthesis of macromolecules | D. Diffusion of oxygen into the cell |
-
- 79.** Homologus structures are evidence of evolution produced from: (1)
- | | |
|-----------------------------|---------------------------|
| A. comparative anatomy | B. Comparative embryology |
| C. Comparative biochemistry | D. Paleontology |
-
- 80.** The normal microbiota population that occupy or live in our bodies temporarily known as (1)
- | | |
|-------------------------|------------------------|
| A. Pathogenic organisms | B. Resident microbiota |
| C. Transient microbiota | D. infectious microbes |
-
- 81.** Fungi show filamentous growth and branches to form a network mass of fungus vegetative body known as (1)
- | | |
|-----------|--------------|
| A. Spore | B. Sporangia |
| C. hyphae | D. mycelium |
-
- 82.** Which of the following organism transmits Africa sleeping sickness Trypanosomiasis? (1)
- | | |
|---------------|-------------|
| A. Sand fly | B. Mosquito |
| C. Tsetse fly | D. Housefly |

-
- 83.** Which group of archaeobacterial are called salt loving bacteria? (1)
- | | |
|-----------------|----------------|
| A. Halophiles | B. Methanogens |
| C. Thermophiles | D. Acidophilus |
-
- 84.** Which of the following is chromosomal mutation? (1)
- | | |
|-----------------|--------------|
| A. Substitution | B. Inversion |
| C. Deletion | D. Addition |
-
- 85.** Hummus & chimpanzees, both evolved from a common ancestor that lived about (1)
- | | |
|-------------------------|-------------------------|
| A. 20 million years ago | B. 15 Million years ago |
| C. 10 million years ago | D. 6 million years ago |
-
- 86.** A larger brain allows humans all accept (1)
- A. run faster & moves in upright posture
 - B. Develop arboreal life
 - C. plan in advance & avoid attack
 - D. Develop and use tools & weapons
-
- 87.** Which of the following is a viral disease that show symptom of the swollen and painful parotid glands & pain in chewing and swallowing? (1)
- | | |
|------------|----------|
| A. Measles | B. Polio |
| C. Mumps | D. AIDS |
-
- 88.** Ammonium compounds are oxidized to nitrate by (1)
- | | |
|-----------------------------|--------------------------|
| A. nitrogen fixing bacteria | B. denitrifying bacteria |
| C. ammonifying bacteria | D. nitrifying bacteria |
-
- 89.** Which of the following protozoan disease causes blindness, mental retardation & multi system organ familiarity? (1)
- | | |
|------------------|------------------|
| A. Toxoplasmosis | B. Leishmaniasis |
| C. Trypanosomes | D. Malaria |
-
- 90.** Which one of the following is not explained by the theory of natural selection? (1)
- A. Over reproduction
 - B. Variation in individual member of the species
 - C. Inheritance of acquired characters
 - D. Survival of the fittest.
-
- 91.** Which one of the following is not correct about lactic acid fermentation (1)
- A. pyruvate converted to lactic acid
 - B. Takes place in muscle cell & bacterial cell
 - C. Oxygen is being used up
 - D. Reduced NAD generate oxidized NAD
-
- 92.** In which of the following features are eukaryotic cells distinguished from prokaryotic cells? (1)
- | | |
|-------------------------------|---|
| A. They have mitochondria | B. They have smaller ribosomes |
| C. Their nuclei lack membrane | D. They lack membrane bounded organelle |

-
- 93.** The use of microbe to extract metals from ores is known as (1)
- | | |
|-----------------------------|-------------------------------|
| A. Bioremediation | B. Bio hydrometallurgy |
| C. Biogas Production | D. Biofuel production |
-
- 94.** Which of the following is not an industrial use of bacteria? (1)
- | | |
|--|---------------------------------|
| A. Recycling of minerals through ecosystems | B. Food processing |
| C. Genetic engineering | D. Waste water treatment |
-
- 95.** In the processes of photosynthesis light is necessary to: (1)
- | | |
|--|---|
| A. Split carbon dioxide | B. Releases energy |
| C. Combine CO ₂ and H ₂ O | D. produces ATP and a reducing substance |

— END OF EXAMINATION —

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